

CO3-AT1-Q2:

Name: D.Uday Kiran

Reg no: 192425161

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Recursion Overhead Analysis in Divide and Conquer Algorithms

1. Problem Statement

Divide and Conquer algorithms such as Binary Search and Merge Sort use recursion to break complex problems into smaller subproblems. Although recursion improves readability and simplifies implementation, it introduces additional memory and execution overhead.

This experiment analyzes recursion overhead by comparing recursive and iterative approaches.

2. Objective

- To study recursion overhead.
- To compare recursive and iterative Binary Search.
- To analyze execution time and stack usage.
- To understand trade-offs between simplicity and efficiency.

3. Algorithms Used

Binary Search

Binary Search divides a sorted dataset into two halves repeatedly.

Recursive approach:

- Function calls itself for smaller sections.
- Uses stack memory for every call.

Iterative approach:

- Uses loops.
- No additional recursion stack required.

4. Recursion Analysis

Recursive Approach

Advantages:

- Simple implementation.
- Easy to understand.
- Matches Divide and Conquer logic.

Disadvantages:

- Requires stack memory.
- Additional function call overhead.
- More memory usage for deep recursion.

Space Complexity:

$O(\log n)$

Iterative Approach

Advantages:

- Less memory usage.
- Faster execution due to no function calls.
- Avoids stack overflow.

Space Complexity:

$O(1)$

5. Comparison

Feature	Recursive Binary Search	Iterative Binary Search
Time Complexity	$O(\log n)$	$O(\log n)$
Stack Usage	More	Less
Execution Speed	Slightly slower	Faster
Code Readability	Easier	Moderate

6. Merge Sort Recursion Analysis

Merge Sort divides the dataset into smaller parts and recursively sorts them.

Advantages:

- Efficient for large datasets.

- Guaranteed $O(n \log n)$ performance.

Disadvantages:

- Requires additional memory.
- Recursive calls increase stack usage.

Space Complexity:

$O(n)$

7. Result Analysis

The experiment shows that recursive algorithms provide better readability and simpler implementation. However, iterative approaches reduce memory consumption and improve execution efficiency.

8. Conclusion

Recursion is useful in Divide and Conquer algorithms because it simplifies problem solving. However, recursion overhead affects memory usage and execution time. Iterative solutions are preferred when system resources are limited, while recursive solutions are preferred for better readability and maintainability.